



LARSEN & TOURBO LIMITED

AUTOMATED PARCEL PACKAGE DAMAGE DETECTION SYSTEM FOR E-COMMERCE FULFILLMENT CENTERS

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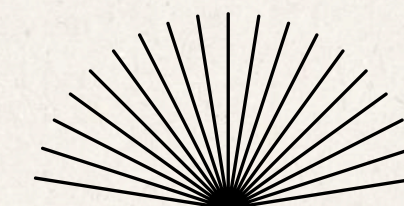




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DATASET COLLECTION PLANNING

THE DATASET WAS PLANNED TO COVER FIVE CORE DAMAGE CLASSES RELEVANT TO CONVEYOR BELT PARCEL INSPECTION. THE GOAL WAS TO COLLECT A DIVERSE SET OF IMAGES THAT REALISTICALLY SIMULATE WAREHOUSE CONVEYOR FOOTAGE.

Planned Damage Classes

Class	Damage Type	Target Annotations
wet	Water damage / broken seals	200+
puncture	Physical puncture damage	200+
dirt	Surface contamination	150+
crushed	Structural crushing	150+
tear_torn	Torn packaging	150+

Collection criteria focused on: overhead/angled camera views, varied lighting conditions, multiple severity levels, and realistic conveyor belt backgrounds.

ROBOFLOW PROJECT SETUP

THE ROBOFLOW WORKSPACE WAS CONFIGURED AS THE CENTRAL PLATFORM FOR DATASET MANAGEMENT, ANNOTATION, VERSIONING, AND MODEL TRAINING.

SETUP CHECKLIST

- ROBOFLOW WORKSPACE CREATED
- NEW PROJECT CREATED: PARCEL DAMAGE DETECTION
- ALL 5 ANNOTATION CLASSES DEFINED
- 195 ORIGINAL IMAGES UPLOADED
- AUTO-ORIENT AND RESIZE (640x640) PREPROCESSING CONFIGURED
- TRAIN / VALIDATION / TEST SPLIT SET TO 70% / 20% / 10%

Dataset Split (Original Images)			
Split	Images	Percentage	Purpose
Train	137	70%	Model learning
Validation	39	20%	Tuning during training
Test	19	10%	Final evaluation

NOTE: AFTER AUGMENTATION, THE TRAINING SET EXPANDS TO ~411 IMAGES WHILE VALIDATION AND TEST SETS REMAIN AT ORIGINAL SIZES. THIS IS STANDARD ROBOFLOW BEHAVIOUR.

IMAGE ANNOTATION — PART 1

BOUNDING BOX ANNOTATIONS WERE APPLIED TO ALL UPLOADED IMAGES USING ROBOFLOW'S ANNOTATION INTERFACE. EACH IMAGE WAS REVIEWED TO ENSURE ALL VISIBLE DAMAGE INSTANCES WERE

Annotation Summary		
Class	Annotations	Status
wet	274	✔ Strong
puncture	205	✔ Good
dirt	160	✔ Moderate
crushed	87	❑ Low
<u>tear torn</u>	66	❑ Very Low
TOTAL	792	195 images

KEY INSIGHT: TOTAL ANNOTATIONS (792) EXCEED IMAGE COUNT (195) BECAUSE SINGLE IMAGES OFTEN CONTAIN MULTIPLE DAMAGE TYPES SIMULTANEOUSLY. ON AVERAGE, EACH IMAGE CONTAINS APPROXIMATELY 4 BOUNDING BOX ANNOTATIONS.

ANNOTATION GUIDELINES APPLIED

- TIGHT BOUNDING BOXES DRAWN AROUND EACH DAMAGE REGION
- MULTIPLE OVERLAPPING DAMAGE TYPES LABELED SEPARATELY PER IMAGE
- AMBIGUOUS CASES (E.G. WET VS. DIRT) RESOLVED BY GROUP CONSENSUS
- ALL 195 IMAGES ANNOTATED WITH AT LEAST ONE BOUNDING BOX

IMAGE ANNOTATION — PART 2 (AUGMENTATION)

TO ADDRESS CLASS IMBALANCE AND DATASET SIZE LIMITATIONS, AUGMENTATION WAS APPLIED VIA ROBOFLOW'S AUGMENTATION PIPELINE AT 3x MULTIPLIER, EXPANDING 195 IMAGES TO 469 TRAINING IMAGES.

Augmentation Settings Applied (Free Tier)		
Augmentation	Setting	Rationale
Rotation	-15° to +15°	Parcels arrive at slight angles on conveyor
Crop	0% to 15%	Simulates partial parcel visibility
Brightness	-25% to +25%	Warehouse lighting variation
Exposure	-15% to +15%	Industrial camera exposure differences
Blur	0 to 1.5px	Simulates motion blur from fast-moving belt
Noise	0% to 2%	Industrial camera sensor noise
Saturation	-20% to +20%	<u>Color</u> variation in damage appearance

AUGMENTATIONS EXCLUDED AND REASONS

- 90° ROTATION — FIXED OVERHEAD CONVEYOR CAMERA; UNREALISTIC ORIENTATION
- GRAYSCALE — COLOR IS CRITICAL FOR DETECTING WET/DIRT DAMAGE CLASSES
- HUE — MAY DISTORT DAMAGE COLOR CUES AND MISLEAD THE MODEL
- SHEAR — TOO GEOMETRICALLY EXTREME FOR FIXED CAMERA ANGLE

Dataset Before vs After Augmentation

Metric	Before (v1)	After (v2)
Total Images	195	469
Training Images	137	~411
Validation Images	39	39
Test Images	19	19
Resolution	—	640 × 640
Multiplier Applied	—	3×

ANNOTATION REVIEW

A PEER REVIEW PROCESS WAS CONDUCTED WITHIN THE TEAM TO ENSURE ANNOTATION QUALITY AND CONSISTENCY BEFORE LOCKING THE DATASET FOR TRAINING.

REVIEW CRITERIA

- **BOUNDING BOXES ARE TIGHT AND CLOSELY FITTED AROUND DAMAGE REGIONS**
- **NO SIGNIFICANT DAMAGE AREAS ARE MISSED OR LEFT UNLABELED**
- **CLASS LABELS ARE CORRECTLY APPLIED (WET VS. DIRT CONFUSION RESOLVED)**
- **IMAGES REPRESENTING CONVEYOR-LIKE ANGLES ARE PRIORITIZED**
- **AUGMENTED IMAGES VISUALLY REVIEWED FOR REALISM**

KNOWN LIMITATIONS IDENTIFIED

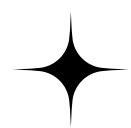
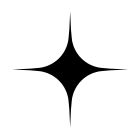
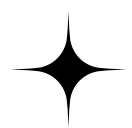
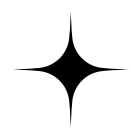
- **CRUSHED (87) AND TEAR_TORN (66) CLASSES REMAIN UNDERREPRESENTED EVEN AFTER AUGMENTATION**
 - **NO SEVERITY LEVEL SUB-LABELING (MILD/MODERATE/SEVERE) APPLIED IN THIS MILESTONE**
 - **SOME IMAGES MAY NOT FULLY REPLICATE INDUSTRIAL CONVEYOR BELT CAMERA ANGLES**
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: Dataset Finalization

The dataset was finalized as Version 2 (v2) in Roboflow, ready for model training. All preprocessing, augmentation, and split settings were locked

Final Dataset Configuration

Parameter	Value
<u>Roboflow</u> Version	v2 (Augmented)
Total Images	469
Train / Val / Test Split	70% / 20% / 10% (on originals)
Image Resolution	640 × 640 <u>px</u>
Auto-Orient	Enabled
Filter Nulls	Enabled
Augmentation Multiplier	3×
Augmentations Applied	7 (Rotation, Crop, Brightness, Exposure, Blur, Noise, Saturation)



Model Training Configuration

Parameter	Value	Reason
Model Architecture	RF-DETR Small	Best accuracy on small datasets
Dataset Version	v2 (469 images)	Augmented for better balance
Training Platform	<u>Roboflow</u> Cloud	Free tier GPU training
Baseline mAP@50	26.9%	First run on 195 original images
Target mAP@50	45–55%	Expected after augmentation retraining

IMPROVEMENT ROADMAP (POST-MILESTONE 2)

- COLLECT 150+ ADDITIONAL IMAGES FOR CRUSHED AND TEAR_TORN CLASSES
- ADD SEVERITY LEVEL ANNOTATIONS (MILD / MODERATE / SEVERE) PER CLASS
- CAPTURE REAL CONVEYOR BELT FOOTAGE WITH OVERHEAD ANGLES AND MOTION BLUR
- RETRAIN ON EXPANDED DATASET TARGETING MAP@50 > 60%
- INTEGRATE TRAINED MODEL WITH POWER BI DASHBOARD FOR MILESTONE 3

SUMMARY

MILESTONE 2 HAS BEEN SUCCESSFULLY COMPLETED. THE DATASET WAS COLLECTED, ANNOTATED, REVIEWED, AUGMENTED, AND FINALIZED IN ROBOFLOW. A BASELINE RF-DETR SMALL MODEL WAS TRAINED ACHIEVING 26.9% MAP@50, WHICH IS EXPECTED FOR AN INITIAL RUN ON 195 IMAGES. RETRAINING ON THE AUGMENTED V2 DATASET (469 IMAGES) IS PROJECTED TO SIGNIFICANTLY IMPROVE PERFORMANCE TOWARD THE 45-55% MAP TARGET. CLASS IMBALANCE IN CRUSHED AND TEAR_TORN REMAINS A KNOWN LIMITATION TO BE ADDRESSED IN SUBSEQUENT MILESTONES.

Milestone Task	Status
Dataset Collection Planning	✔ Complete
<u>Roboflow</u> Project Setup	✔ Complete
Image Annotation Part 1	✔ Complete
Image Annotation Part 2 (Augmentation)	✔ Complete
Annotation Review	✔ Complete
Dataset Finalization	✔ Complete
Baseline Model Training	✔ Complete (<u>mAP</u> : 26.9%)
Augmented Model Retraining	❑ In Progress